Option 1: Comparative scRNA-seq Analysis of Gene Expression Differences Between Lung and Trachea Populations (Tabula Muris FACS, Smart-seq2)

Introduction

Single-cell RNA sequencing (scRNA-seq) enables analysis of transcriptional variation across individual cells [1]. In Assignments 3 and 4, we used scRNA-seq data from the Tabula Muris project to explore dimensionality reduction, clustering, and differential expression in myeloid and non-myeloid tissues. In this project, we extend those methods to two closely related respiratory tissues, the mouse lung and trachea, from the Tabula Muris FACS Smart-seq2 dataset. Similar to Assignments 3 and 4, we processed the scRNA-seq data by filtering low-quality cells, normalizing library size, applying a square-root transformation, performing PCA for dimensionality reduction [2], and using UMAP for visualization [3]. Unsupervised clustering was performed using Phenograph, K-means, and Spectral methods, followed by differential expression analysis to identify genes that distinguish clusters [4][5]. The lung supports gas exchange, containing alveolar type I and II epithelial cells, resident macrophages, and extensive microvascular endothelium [6]. In contrast, the trachea is a conducting airway rich in multiciliated and basal epithelial cells supporting mucociliary clearance [7]. These functional differences imply distinct regulatory programs, predicting transcriptional separation between surfactant-producing lung epithelium and mucociliary tracheal epithelium [6][7].

Motivation and Dataset Selection

The Tabula Muris FACS Smart-seq2 dataset offers high-quality single-cell transcriptomes with curated metadata for many mouse tissues [8]. I selected lung and trachea because (1) they are biologically related respiratory tissues, (2) the dataset includes complete and well-annotated count matrices for both, and (3) they contain overlapping broad cell classes (e.g., epithelial, endothelial, stromal, and immune cells) but differ in physiological roles and microenvironments. Additionally, the presence of well-established subtissue markers such as EPCAM (epithelial) and Endomucin (endothelial) enables more direct evaluation of whether known biological distinctions appear naturally in the low-dimensional structure [9][10].

Our goal is to determine how differences in transcriptional programs and cell composition between lung and trachea appear in low-dimensional structure, and which marker genes drive tissue- and lineage-specific separations. This motivates the central question of the study: ***How do differences in transcriptional programs and cellular composition between lung and trachea manifest in low-dimensional structure, and which gene signatures underpin these tissue-specific and lineage-specific separations?*** This question is interesting because it allows us to evaluate (1) whether tissue identity is detectable in expression space, (2) how strongly subtissue markers contribute to the major axes of variation, and (3) whether cluster-specific genes identified through differential expression correspond to known biological distinctions or suggest novel patterns.

Methods and Dataset

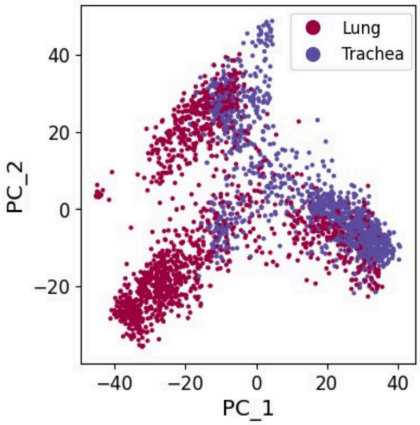
The Tabula Muris Consortium (2018) produced a single-cell transcriptomic atlas of 20 mouse organs and tissues using FACS-sorted Smart-seq2 and droplet-based 10x Genomics [8]. The study highlights how high-quality scRNA-seq across diverse tissues reveals cellular heterogeneity, lineage relationships, and

tissue-specific expression programs. With curated metadata, standardized QC, and expert cell-type annotations, Tabula Muris provides a benchmark for cross-tissue comparisons and for validating whether computational clusters correspond to biologically meaningful cell types.

We analyzed two FACS-sorted Smart-seq2 count matrices from the Tabula Muris dataset (Lung-counts.csv and Trachea-counts.csv) together with metadata (metadata_FACS.csv) [1]. The HW3 and HW4 preprocessing pipeline was replicated exactly. Lung (1,923 cells × 23,433 genes) and trachea (1,391 cells × 23,433 genes) matrices were concatenated into a combined dataset (3,314 cells × 23,433 genes), aligned with metadata fields using the plate barcode and mouse ID extracted from the cell indices. It includes tissue, subtissue, mouse.id, mouse.sex, plate.barcode, and FACS.selection. Cells with library sizes outside the 10th-90th percentile range are removed to exclude potential outliers (e.g., low-quality cells or doublets). After removing genes expressed in fewer than 10 cells, the final processed dataset contained 2,650 cells × 16,619 genes.

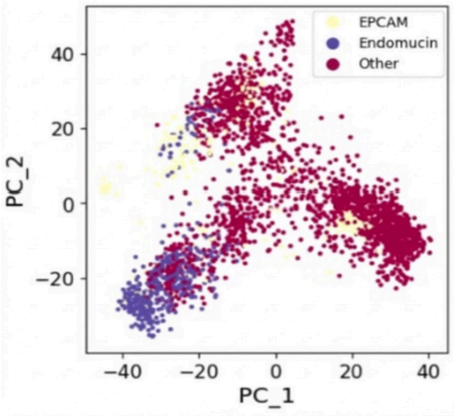
Preprocessing steps included library-size normalization and square-root transformation for variance stabilization. Dimensionality reduction followed HW3: we computed a 50-component PCA using dense PCA and generated UMAP embeddings from the top PCs for visualization. Clustering replicated HW4: Phenograph, K-means (with prediction strength to select k, optimal at $k = 4$), and spectral clustering. Differential expression was performed using t-tests comparing each cluster to all other cells, ranking marker genes by absolute t-statistic [11]. Global structure genes and tissue-specific markers were examined across PCA, UMAP, and cluster-level comparisons.

PCA of Lung and Trachea Cells, Colored by Tissue



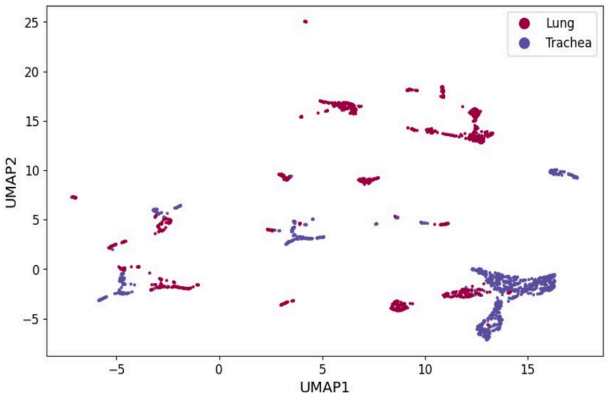
PCA shows clear PC1/PC2 separation between lung and trachea cells. Tissue identity is a major source of variation, reflecting differences in composition or transcriptional

PCA of Lung and Trachea Cells, Colored by Subtissue Marker



PCA colored by EPCAM (epithelial) vs Endomucin (endothelial) shows stronger separation than tissue identity. The two marker-defined lineages occupy distinct PC

UMAP Embedding of Lung and Trachea Cells, Colored by Tissue



UMAP shows two major manifolds dominated by lung and trachea, with overlap from shared cell types. Compared to PCA, UMAP better resolves nonlinear boundaries between epithelial, endothelial, and stromal

programs. Some overlap likely reflects shared epithelial/stromal lineages, but each tissue forms a distinct cluster.	*regions, indicating lineage explains more variance than lung vs trachea origin.*	*compartments [14]. Lung-specific regions appear more dispersed, suggesting greater immune or stromal diversity.*
Limitation: PCA captures only linear structure, so nonlinear patterns and subtle subpopulations (especially within shared epithelial or endothelial lineages) may be missed.	**Limitation:** Subtissue labels based on only two markers can oversimplify heterogeneity. More markers would give higher-resolution identities.	**Limitation:** UMAP is sensitive to initialization, neighbor parameters, and random seed, so embedding shape can shift between runs. Interpret biology from stable global patterns, not fine-grained layout.

Results

1. Tissue identity is a dominant driver of variation

PCA and UMAP separate lung and trachea cells into distinct regions, likely reflecting differences in epithelial composition. Lung-enriched regions correspond to surfactant-producing alveolar type II cells (Sftpc) [12], while trachea-enriched regions align with basal and ciliated lineages expressing Krt5 [7] and Foxj1 [13]. This illustrates that unsupervised structure reflects meaningful physiological specialization.

Limitation: Some apparent tissue separation may reflect sampling or technical effects (e.g., plate or mouse differences) rather than purely biological signal.

2. Subtissue identity explains more variation than tissue identity

EPCAM-positive epithelial and Endomucin-positive endothelial cells separate more strongly than tissue labels, indicating lineage programs are the dominant axis of variation. Tissue identity refines these lineage-level programs, consistent with developmental branching followed by organ-specific specialization.

Limitation: Using only two markers oversimplifies cellular diversity and does not annotate immune, rare, or intermediate populations.

3. UMAP reveals shared vs tissue-specific manifolds

UMAP shows overlapping epithelial and endothelial states across tissues, but tissue-specific extensions likely represent adaptation to environment — surfactant production in lung vs mucosal defense in trachea. Thus, conserved cell modules coexist with localized specialization.

Limitation: UMAP geometry is optimized for visualization and is sensitive to parameter choices; distances and fine-scale structure should not be interpreted quantitatively.

Overall Takeaway: These findings demonstrate that unsupervised workflows recover both shared respiratory cell modules and tissue-specific specialization. This supports the biological expectation that related organs share core lineages but diverge through environment-dependent regulatory programs.

References

[1] Kolodziejczyk, A. A., Kim, J. K., Svensson, V., Marioni, J. C., & Teichmann, S. A. (2015). *The technology and biology of single-cell RNA sequencing*. Molecular Cell, 58(4), 610–620. https://doi.org/10.1016/j.molcel.2015.04.005. Accessed 9 December 2025.

[2] Ringnér, M. (2008). *What is principal component analysis?* Nature Biotechnology, 26(3), 303–304. https://doi.org/10.1038/nbt0308-303. Accessed 9 December 2025.

[3] McInnes, L., Healy, J., & Melville, J. (2018). *UMAP: Uniform Manifold Approximation and Projection*. Journal of Open Source Software, 3(29), 861. https://doi.org/10.21105/joss.00861. Accessed 9 December 2025.

[4] Levine, J. H., Simonds, E. F., Bendall, S. C., Davis, K. L., Amir, E. D., Tadmor, M. D., Litvin, O., Fienberg, H. G., Jager, A., Zunder, E. R., Finck, R., Gedman, A. L., Radtke, I., Downing, J. R., Pe'er, D., & Nolan, G. P. (2015). *Data-driven phenotypic dissection of AML reveals progenitor-like cells that correlate with prognosis*. Cell, 162(1), 184–197. https://doi.org/10.1016/j.cell.2015.05.047. Accessed 9 December 2025.

[5] von Luxburg, U. (2007). *A tutorial on spectral clustering*. Statistics and Computing, 17(4), 395–416. https://doi.org/10.1007/s11222-007-9033-z. Accessed 9 December 2025.

[6] Brandt, J. P., & Mandiga, P. (2023). *Histology, Alveolar Cells*. In StatPearls [Internet]. StatPearls Publishing. Available from: https://www.ncbi.nlm.nih.gov/books/NBK557542/. Accessed 9 December 2025.

[7] Rock, J. R., Onaitis, M. W., Rawlins, E. L., Lu, Y., Clark, C. P., Xue, Y., Randell, S. H., & Hogan, B. L. M. (2009). *Basal cells as stem cells of the mouse trachea and human airway epithelium*. Proceedings of the National Academy of Sciences, 106(31), 12771–12775. https://doi.org/10.1073/pnas.0906850106. Accessed 9 December 2025.

[8] The Tabula Muris Consortium., Overall coordination., Logistical coordination. *et al.* Single-cell transcriptomics of 20 mouse organs creates a *Tabula Muris*. *Nature* 562, 367–372 (2018). https://doi.org/10.1038/s41586-018-0590-4. Accessed 5 December 2025.

[9] Sinha, M., & Lowell, C. A. (2016). *Immune defense protein expression in highly purified mouse lung epithelial cells*. American Journal of Respiratory Cell and Molecular Biology, 54(6), 802–813. https://doi.org/10.1165/rcmb.2015-0171OC. Accessed 9 December 2025.

[10] Liu, C., Shao, Z. M., Zhang, L., Beatty, P., Sartippour, M., Lane, T., Livingston, E., & Nguyen, M. (2001). *Human endomucin is an endothelial marker*. Biochemical and Biophysical Research Communications, 288(1), 129–136. https://doi.org/10.1006/bbrc.2001.5737. Accessed 9 December 2025.

[11] Student. (1908). *The probable error of a mean*. Biometrika, 6(1), 1–25. Accessed 9 December 2025.

[12] Barkauskas, C. E., Cronce, M. J., Rackley, C. R., Bowie, E. J., Keene, D. R., Stripp, B. R., Randell, S. H., Noble, P. W., & Hogan, B. L. M. (2013). *Type 2 alveolar cells are stem cells in adult lung*. Journal of Clinical Investigation, 123(7), 3025–3036. https://doi.org/10.1172/JCI68782. Accessed 9 December 2025.

[13] Blatt, E. N., Yan, X. H., Wuerffel, M. K., Hamilos, D. L., & Brody, S. L. (1999). *Forkhead transcription factor HFH-4 expression is temporally related to ciliogenesis*. American Journal of Respiratory Cell and Molecular Biology, 21(2), 168–176. https://doi.org/10.1165/ajrcmb.21.2.3691. Accessed 9 December 2025.

[14] Villares, M., Saunders, C. M., & Fey, N. (2024). *Comparison of dimensionality reduction techniques for the visualisation of chemical space in organometallic catalysis*. Artificial Intelligence in Chemistry, 1(1), 100013. https://doi.org/10.1016/j.aichem.2024.100055. Accessed 9 December 2025.